Network Intrusion Detection Using Multi-Agent Autonomous AI system

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Introduction

Cybersecurity threats continue to increase in both scale and sophistication, making it difficult for traditional intrusion detection systems to provide reliable and interpretable results. Most existing systems rely either on machine learning models that lack explainability or on static rule-based systems that lack adaptability.

This project introduces a **multi-agent cybersecurity system** that integrates machine learning, rule-based reasoning, and AI-based decision-making to simulate the behavior of a real-world Security Operations Center (SOC). The system is designed to detect, analyze, and respond to network threats in a structured and explainable manner.

Objective

The main objectives of the system are:

- Detect malicious network traffic
- Classify attack types such as DoS, DDoS, and PortScan
- Provide interpretable reasoning behind detections
- Recommend appropriate response strategies
- Demonstrate a scalable multi-agent cybersecurity architecture

Core Idea

Instead of relying on a single detection model, the system uses multiple agents that collaborate in a pipeline. Each agent performs a specific role, allowing the system to move from simple classification to structured reasoning and system-level decision-making.

This approach improves:

- Interpretability of results
- Scalability of the system
- Accuracy of decisions

System Architecture:

Network Traffic Input

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Detection Agent (Machine Learning)

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Analysis Agent (Rule-Based Reasoning)

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Threat Prioritization

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Aggregated Summary

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Research Agent (LLM)

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Response Agent (LLM)

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Dashboard Visualization

Methodology

Data Preparation and Network Stream Simulation

The system uses a cleaned version of the CICIDS2017 dataset, which contains labeled network flow records representing both benign and malicious activity.

Key preprocessing steps include:

- Removal of irrelevant attributes such as IP addresses and timestamps
- Handling of missing or infinite values
- Normalization of feature values
- Simplification of attack labels

Network Stream Simulation

To demonstrate real-time behavior, a synthetic data generator was developed. Instead of using static data, the system simulates a network stream by sampling from real distributions and injecting attack patterns.

The generated traffic includes:

- Normal traffic representing legitimate user behavior
- DoS traffic with moderate packet rate increases
- DDoS traffic with extreme packet and byte rate spikes
- PortScan traffic with asymmetric communication patterns

Attack Behavior Characteristics

- **DoS Attack**
  - High packet rate
  - Moderate increase in data volume
  - Continuous flow
- **DDoS Attack**
  - Extremely high packets per second
  - Large data throughput
  - High SYN flag activity
  - Short flow durations
- **PortScan Attack**
  - Many forward packets with little response
  - High SYN count
  - Low packet size
  - Asymmetric traffic patterns

These patterns allow the system to realistically simulate attack scenarios.

Detection Agent (**Machine Learning**)

The Detection Agent uses a Random Forest classifier trained on network flow data.

The agent performs:

- Feature validation to ensure required inputs exist
- Feature alignment to match model structure
- Prediction of traffic type
- Confidence scoring
- Risk classification

Output Structure

status: ATTACK or NORMAL

prediction: attack type

confidence: probability score

risk_level: HIGH, MEDIUM, or LOW

Risk Classification

- Probability > 0.8 → HIGH
- 0.5 – 0.8 → MEDIUM
- < 0.5 → LOW
- Benign traffic → LOW

Sample Feature Set Used by the Model

Feature	Description
Flow Duration	Total duration of the network flow
Total Fwd Packets	Packets sent forward
Total Backward Packets	Packets sent backward
Total Length of Fwd Packets	Size of forward packets
Total Length of Bwd Packets	Size of backward packets
Flow Bytes/s	Bytes per second
Flow Packets/s	Packets per second

Flow IAT Mean	Average time between packets
Fwd IAT Mean	Forward packet interval
Bwd IAT Mean	Backward packet interval
Packet Length Mean	Average packet size
SYN Flag Count	Number of SYN packets
ACK Flag Count	Number of ACK packets

These features capture traffic behavior, intensity, and communication patterns.

Analysis Agent (Rule-Based Reasoning)

The Analysis Agent enhances detection using domain knowledge.

It evaluates:

- SYN flag activity
- Packet rate
- Data volume
- Traffic asymmetry

Example Rules

- High SYN + low ACK → SYN flood attack
- High packets/s → DoS or DDoS
- High bytes/s → abnormal data transfer
- Asymmetry → PortScan

Risk Scoring

Each rule contributes to a cumulative score:

Risk Score = Total Score / 9

Output

- Attack type
- Risk score
- Confidence level
- Explanation
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Multi-Agent AI System (**Research** + **Response Agents**)

The system uses CrewAI to implement higher-level reasoning.

-Research Agent

- Explains attack behavior
- Provides context
- Determines severity

-Response Agent

- Recommends action
- Suggests mitigation steps
- Outputs structured response

These agents operate only on **top-priority threats**.

-System Optimization

The system was optimized using:

- Separation of detection and reasoning layers
- Storage of results to prevent recomputation
- Selection of top threats for AI processing
- Aggregation of inputs for single AI calls

This ensures:

- Faster execution
- Consistent outputs
- Reduced computational cost

8. Dashboard and Integration

The dashboard provides real-time monitoring and analysis.

Features

- Live traffic visualization
- Detection and analysis outputs
- Risk metrics
- Alert system
- AI diagnosis page

Workflow

1. Detection agent processes traffic
2. Analysis agent evaluates behavior
3. Results are stored

4. AI agents analyze top threats
5. Dashboard displays outputs

9. Team Contributions

Member	Role	Contribution
Justin Silvere	Project Manager	ML model, system integration
Jack Hayes	AI Developer	Research and response agents
Morsey Rosales	Data Engineer	Dataset preparation
Jason Sin	Analysis Developer	Feature reasoning and validation
Barry Lin	Dashboard Developer	UI and visualization

10. Advantages

- Combines ML and rule-based reasoning
- Provides explainable outputs
- Efficient multi-agent design
- Realistic SOC workflow
- Scalable architecture

11. Limitations

- Synthetic network attack data stream may not fully represent real-world network traffic
- Threshold values require tuning
- Model performance depends on dataset quality
- LLM responses depend on prompt design

12. Areas for Improvement

The system can be improved by:

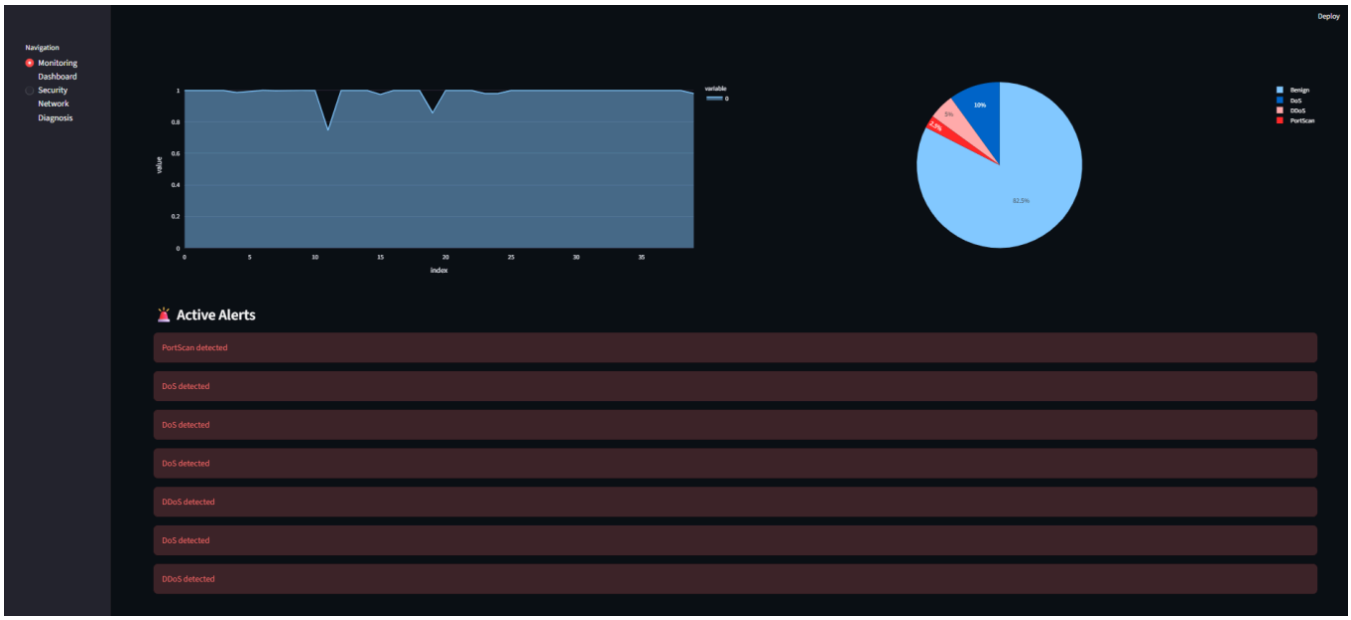
- Integrating real-time packet capture instead of simulated data
- Using adaptive thresholds instead of static ones
- Incorporating deep learning models
- Automating response actions (firewall integration)
- Improving attack diversity in generated data

13. Conclusion

Overall, this semester project demonstrates a complete multi-agent cybersecurity system capable of detecting, analyzing, and responding to network threats.

By combining Machine Learning, rule-based reasoning, and Autonomous AI agents that drive decision-making, the system provides a scalable and explainable approach to modern cybersecurity challenges.

Dashboard /Output:



Security Network Diagnosis using Multi-Agents

Threat Priority Table

	Threat	Risk Score	Confidence	Severity
0	DoS	0.56	0.975	MEDIUM
1	DoS	0.33	0.747	MEDIUM
2	DoS	0.33	0.858	MEDIUM

System Insight

The analyzed data indicates a series of 'Denial of Service' (DoS) attacks on the network.

DoS attacks are hazardous as they disrupt network service or connectivity that leads to loss of access to network resources, predominantly affecting the availability aspect of the CIA (Confidentiality, Integrity, Availability) triad in information security.

Key Indicators of a DoS attack include unusual high network traffic, sluggish network performance, unavailability of a particular website, and an increase in the number of spam emails received.

The severity of these attacks varies as per the risk score and confidence; the higher these numbers are, the more severe the attack is.

Here are three instances of DoS attacks:

1. The first attack has a risk score of 0.56 with the confidence level of 0.975, implying it's a significant threat due to its higher risk score and confidence level.
2. The second and the third attacks both carry a risk score of 0.33, but their confidence levels vary as 0.747 and 0.858, respectively. These attacks are less severe compare to the first one, but still, they carry considerable risk considering the high confidence levels.

In summary, DoS attacks identified in your network are serious threats disrupting the availability of services and should be addressed immediately to mitigate further interruption in services.

Recommended Actions

Action: Block

Reason: The attacks identified have a high-risk score and confidence levels indicating a Denial-of-Service (DoS) attack. These attacks can over-flood network resources, causing potential availability issues to legitimate users. Given such a significant threat, our best course of action is to block these attacks immediately to minimize their impact on our network and systems.

Response Plan:

1. Immediately block the suspicious IP addresses identified in the threat detection from accessing our network. This can be done by adjusting the firewall rules to disallow incoming traffic from these IPs.
2. Establish rate limiting on the network to control the amount of incoming traffic and safeguard the system from becoming overwhelmed.
3. Engage network traffic analysis by utilizing advanced tools that can help us better identify and mitigate future DoS attacks.
4. Conduct a thorough system vulnerability check to ensure no weaknesses were exploited and apply necessary patches wherever deficiencies are identified.
5. Start an investigation to ascertain how these attacks materialized, and determine if any data breaches occurred or system damages were inflicted.
6. Based on the investigation results, enhance the existing network and system security measures to prevent a recurrence of such incidents.
7. Communicate the incident, actions taken, and plans moving forward clearly to all relevant stakeholders.
8. Finally, monitor the network continuously to detect any unusual activities or potential recurrence of attacks.

This comprehensive approach will ensure we stave off the current threat and improve our overall readiness for any similar future threats.